AMITY INSTITUTE OF INFORMATION TECHNOLOGY

WEEKLY PROGRESS REPORT
FOR THE WEEK COMMENCING (08/06/2026-14/06/2026)

WPR No: 04
Program: MCA

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ENROLLMENT NO.- A010145025215

SEMESTER: 3rd
FACULTY GUIDE NAME: Dr. Garima Bohra

PROJECT TITLE: RADMAP – GPS Enabled Radiation Mapping Prototype using Raspberry Pi

TARGETS FOR THE WEEK:

1. Researched radiation detection systems used in nuclear laboratory environments.
2. Studied hardware components: LiDAR Sensor, SG90 Servo Motor, and ESP8266 camera module.
3. Designed and developed a monitoring dashboard using Streamlit.
4. Explored integration of temperature and humidity sensors for environmental monitoring.
5. Planned architecture for radiation data collection and 3D mapping workflow.

ACHIEVEMENTS OF THE WEEK:

1. Completed literature review on radiation monitoring and hazardous environment systems.
2. Finalized hardware selection for prototype development.
3. Successfully built the initial Streamlit dashboard for sensor visualization.
4. Designed end-to-end system architecture for multi-sensor integration.

5. Established the concept of real-time 3D Digital Twin radiation mapping.

FUTURE WORK PLANS:

1. **Hardware Integration with Raspberry Pi** Integrate all core hardware components including Raspberry Pi, LiDAR Sensor, SG90 Servo Motor, and ESP8266 camera module into a unified system architecture.
2. **Radiation Detection Module Implementation** Connect and calibrate the radiation detector sensor to accurately capture radiation intensity levels and validate readings through controlled testing environments.
3. **Multi-Sensor Data Acquisition & Fusion** Develop a data pipeline to collect and synchronize radiation, temperature, humidity, distance scanning, and camera feed data for combined environmental analysis.
4. **Real-Time Dashboard Communication System** Establish wireless/API-based communication between hardware modules and the Streamlit dashboard for continuous real-time monitoring, storage, and visualization of sensor data.
5. **3D Digital Twin Radiation Mapping Development** Build a 3D visualization engine that converts collected sensor data into a digital twin model to represent radiation distribution zones, hotspot detection, and anomaly prediction using AI/ML techniques.